AMAZON ANALYSIS USING SQL

Amazon, a global leader in e-commerce, has achieved significant success in markets like the U.S., Europe, and Asia. In Brazil, Amazon connects small and medium businesses with millions of customers, becoming a key player. Given the similarities between Brazil and India—such as large populations and diverse consumer bases—there's an opportunity to replicate success in India. This analysis will help Amazon India make informed decisions, enhance customer experience, and seize new opportunities.

Analysis-I

1.Average payment value- Standardizing average payment amounts help amazon simplify financial reporting and comparisons across Payment types

```sql
select payment_type,
      round( Avg(payment_value),0) AS rounded_average_payment
from amazon_analysis.payments
group by payment_type
order by  rounded_average_payment asc;
```

Data Output Messages Notifications

	payment_type character varying	rounded_average_payment numeric
1	not_defined	0
2	voucher	66
3	debit_card	143
4	boleto	145
5	credit_card	163

Analysis:

- Rounding off payment values simplifies financial reporting and avoids unnecessary complexity with decimal
- Higher value purchases are made through credit card, whereas vouchers seem to be used for any small payments

Recommendation:

- A further analysis on the "not defined" category can be done to understand why payments are not categorized. The average payment value is 0 suggesting that It could be due to incomplete /error driven data entries.

2. **Distribution of orders by payment types** - Knowing the percentage share of each payment type allows amazon to refine its payment strategies and promote the most popular method

```sql
select payment_type,
       round(count(*)*100/sum(count(*))over(),1) as percentage_orders
from amazon_analysis.payments
group by payment_type
order by percentage_orders desc;
```

Data Output Messages Notifications

	payment_type character varying	percentage_orders numeric
1	credit_card	73.9
2	boleto	19.0
3	voucher	5.6
4	debit_card	1.5
5	not_defined	0.0

Analysis-

- The percentage distribution shows credit card is the highest value mode of payment for the majority of population

Recommendations-

- Amazon India can utilize partnership, Discounts, cashback for credit card users to drive more revenue and profit for the organization. Offers, instant cashback for debit card payments can drive more sales

3. Targeted promotions by price range – identifying “smart” products within a specific price range enables amazon to design promotions for mid-range customers

```sql
select o.product_id,
       round(o.price,0)As price
FROM amazon_analysis.order_items o
JOIN amazon_analysis.product p
ON o.product_id=p.product_id
where o.price between 100 and 500
AND p.product_category_name like '%smart%'
order by o.price desc;
```

Data Output Messages Notifications

	product_id character varying	price numeric
1	1df1a2df8ad2b9d3aa49fd851e314...	440
2	7debe59b10825e89c1cbcc8b190c...	350
3	ca86b9fe16e12de698c955aedff0ae...	349
4	ca86b9fe16e12de698c955aedff0ae...	349
5	0e52955ca8143bd179b311cc454a...	335
6	7aeaa8f3e592e380c420e8910a717...	330
7	7aeaa8f3e592e380c420e8910a717...	330

Analysis:

- The mid zone (300-400) products are at consistent demand, and budget zone (100-300) are highest repeat products and lowest price range products

Recommendations:

- The BRL 100-500 band = roughly **₹1,000-₹5,000** this is exactly sweet spot for smart home/accessories in India
- The (400-500) premium gap – amazon India can push premium smart devices with Emis to fill this gap

4. Seasonal sales patterns- Finding the top performing months helps amazon to plan inventory and marketing during the peak seasonal patterns

```sql
select to_char( order_purchase_timestamp,'month') as month,
       round(sum(oi.price), 0) as total_sales
from amazon_analysis.orders o
join amazon_analysis.order_items oi
on o.order_id =oi.order_id
group by to_char(order_purchase_timestamp,'month')
order by total_sales desc
limit 3;
```

Data Output Messages Notifications

	month text	total_sales numeric
1	may	1502589
2	august	1428658
3	july	1393539

Analysis:

- May is the peak season that drives sales certainly due to summer which corelates with high demand of cooling appliances (Acs, coolers) followed by Q3 (July& august) fuelled by Independence Day sales and early “back-to-college” shopping

Recommendations:

- inventory pre-allocation- planning inventory buildup and readiness at least 4-5 weeks before these months hit. Launching targeted promotions & Marketing in these windows will more likely to get highest ROAS (return on ad spent) and high profit conversion intent

5. price variation by category- Analysing various categories of products with the price variation helps amazon with marketing pricing and promotional strategies

```sql
select p.product_category_name,round(max(o.price)-min(o.price),0) as price_difference
from amazon_analysis.order_items o
join amazon_analysis.product p
on p.product_id=o.product_id
WHERE p.product_category_name IS NOT NULL
group by product_category_name
having max(o.price)-min(o.price)>500
order by price_difference desc ;
```

Data Output Messages Notifications

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	product_category_name character varying	price_difference numeric
1	utilidades_domesticas	6732
2	pcs	6695
3	artes	6496
4	eletroportateis	4793
5	instrumentos_musicais	4395
6	consoles_games	4095
7	esporte_lazer	4055

Analysis: The massive gap tells us that inventory spans from entry/budget level to premium/luxury buyers, the categories such as “utilidades domesticas”(6732 BRL difference), “pcs”(6695BRL) exhibit very large gap indicating they include both budget-friendly and premium products

Recommendations: sub-categorize product into “budget”, standard and premium sub-tags this improves filter accuracy for users and

- using target marketing high spread-using aspirational videos and
- low spread – using value-based marketing

6. Enhancing customer experience – Analyzing customer payments types shows which methods are most usable for finding consistent transaction amount

```sql
select payment_type,
round(stddev(payment_value),2) as std_deviation
from amazon_analysis.payments
where payment_type!='not_defined'
group by payment_type
order by std_deviation asc;
```

Data Output Messages Notifications

	payment_type character varying	std_deviation numeric
1	voucher	115.52
2	boleto	213.58
3	credit_card	222.12
4	debit_card	245.79

Analysis: lower stdev=more consistent and predictable transactions , vouchers being most consistent makes sense – vouchers are basically fixed denomination (50,100,200BRL), so amounts don't vary wildly
Recommendations –

- Voucher/coupon design in india – issue fixed-value coupons (₹100, ₹200, ₹500) debit card variance is too high means inconsistent customer behaviour.
- amazon india can look into different payment options like UPIs, net banking and more to enhance smooth payment modes

7. Fixing incomplete order names – amazon india wants to fix their order names from products where product names are null, not listed properly or incomplete

```
select product_id,product_category_name
from amazon_analysis.product
where product_category_name is null or length(product_category_name)<=1
```

Data Output Messages Notifications

	product_id [PK] character varying	product_category_name character varying
1	a41e356c76fab66334f36de622ecbd...	[null]
2	d8dee61c2034d6d075997acef1870...	[null]
3	56139431d72cd51f19eb9f7dae4d16...	[null]
4	46b48281eb6d663ced748f324108c...	[null]
5	5fb61f482620cb672f5e586bb132ea...	[null]
6	e10758160da97891c2fdcbc35f0f03...	[null]
7	39e3b9b12cd0bf8ee681bbc1c130fe...	[null]

Analysis:

- There are 614 rows of null data which is invisible in search and filters,
- This is a direct revenue leak- products exists but are never discovered, single character entries suggest sellers may be submitting incomplete listings

Recommendations: Make product category a mandatory non-null field, build an automated data quality check. This is a priority task as clean categorization directly impacts with search ranking and conversion

ANALYSIS-II

1. payment types by order value- counting different payment types based of the order value segment helps amazon analyze usage percentage of payment methods over order values

```sql
select
       case when payment_value <200 then 'low'
       when payment_value between 200 and 1000 then 'medium'
       else 'high'
       end as order_value_segment,
       payment_type,
       count(payment_value) as count
from amazon_analysis.payments
group by order_value_segment,payment_type
order by count desc;
```

Data Output Messages Notifications

	order_value_segment text	payment_type character varying	count bigint
1	low	credit_card	60548
2	low	boleto	16444
3	medium	credit_card	15303
4	low	voucher	5476
5	medium	boleto	3162
6	low	debit_card	1287
7	high	credit_card	944

Analysis- credit card dominates every single payment type, boleto is strong in low and medium segments, vouchers are negligible across all segments

Recommendations:

- low segment under (₹2,000)- this is where volume lives optimize checkout speed and payment success rates here. Include cash on delivery (COD), UPI/wallet discounts for low-ticket orders
- Focus on “No-cost Emi”, Buy “now Pay later” options for credit card in high order value segment to increase the average of 944 count by making the monthly payments small

2. price range & average price by category – compare product categories based on minimum, maximum and average prices

```sql
SELECT p.product_category_name,
       MIN(o.price) as min_price,
       MAX(o.price) as max_price,
       round(AVG(o.price),2) as avg_price
FROM amazon_analysis.product p
JOIN amazon_analysis.order_items o
ON p.product_id=o.product_id
GROUP BY product_category_name
ORDER BY avg_price desc;
```

Data Output Messages Notifications

	product_category_name character varying	min_price numeric	max_price numeric	avg_price numeric
1	pcs	34.5	6729	1098.34
2	portateis_casa_forno_e_cafe	10.19	2899	624.29
3	eletrodomesticos_2	13.9	2350	476.12
4	agro_industria_e_comercio	12.99	2990	341.66
5	instrumentos_musicais	4.9	4399.87	281.62
6	eletroportateis	6.5	4799	280.78
7	portateis_cozinha_e_preparadores_de_aliment...	17.42	1099	264.57

Analysis:

- Fashion and food categories cluster at the bottom (25–90 BRL avg) high volume and low margin territory
- Fashion sitting at the bottom avg prices wise but being a massive category it's a volume game with margins

Recommendation:

- Tiered navigation – "shop by price tier" (budget, pro, premium) to match customer intent more accurately
  highlight premium/luxury items in marketing campaigns
- Premium bundling: create bundles that pair high-margin accessories (low price) with high value systems (max price) to increase average order value (AOV)

3. customers with multiple orders – finding customers who placed more than one order overtime

```sql
select c.customer_unique_id,count(o.order_id) as total_orders
from amazon_analysis.customers c
join amazon_analysis.orders o
on c.customer_id=o.customer_id
group by c.customer_unique_id
having count(o.order_id)>1
order by total_orders desc;
```

Data Output Messages Notifications

	customer_unique_id character varying	total_orders bigint
1	a91e80fbe80ddc07de66a5cf927029...	16
2	a6168cd79131e64acef92e3c74d6cc...	16
3	363f980585bf04c1a88fdb986011c5...	16
4	cbd0350d4ccba9772e8e768d4a4a5...	16
5	417b909c0962b2610f1cfeb1c14789...	16
6	5f94af52aef02c968a2e0f01f430864e	16
7	1b6d29725255a77667a8c639eeb4c...	16

Analysis:

- Repeat purchase behaviour is weak overall most customers buy once and don't return
- Returning customers (>1 order) generate more revenue
- Low repeat rate signals either poor retention strategy or customers switching to other platforms

Recommendations:

- Customers having total orders (>10 orders) are our VIP seeds analyze what their buying behaviour and build a loyality program modelled on their behaviour
- Analyze where retention is dropping, low repeat rate =amazon india's biggest opportunity. increase post purchase engagement (review requests, ratings, personalized recommendations)

4. categorizing customers based on orders – categorizing customers based on their orders helps amazon for further analysis and increasing revenue

```sql
with new_cte as(
select customer_id,count(order_id) as total_orders,
case when count(order_id)=1 then 'new'
when count(order_id) between 2 and 4 then'returning'
else 'loyal'
end as customer_type
from amazon_analysis.orders
group by customer_id
)
select c.customer_unique_id,n.customer_type
from new_cte n
join amazon_analysis.customers c
on c.customer_id=n.customer_id;
```

Data Output Messages Notifications

Showing rows: 1 to 10

	customer_unique_id character varying	customer_type text
1	248ffe10d632bebe4f7267f1f44844c9	new
2	b0015e09bb4b6e47c52844fab5fb66...	new
3	4893ad4ea28b2c5b3ddf4e82e79db9...	new
4	0b83f73b19c2019e182fd552c048a2...	new
5	104bdb7e6a6cdceaa88c3ea5fa6b2b...	new
6	14843983d4a159080f6afe4b7f346e7c	new
7	15090f48004f3b0fc18a167ef82af4db	new

Analysis:

- Vast majority are new customers (1 order), with tiny returning customers.
- Loyal customers (5+ orders) contribute disproportionally to revenu0065

Recommendations:

- Amazon Brazil is running acquisition heavy& retention weak business Don't replicate Brazil mistake, balance acquisition spend with retention from day one.
- Target new ->returning customer, send personalized offers within 7 days to push customer into "returning" territory

5. top revenue generating categories – calculating revenue to Analyzing which product categories generate most revenue

```sql
select p.product_category_name,sum(o.price) as total_revenue
from amazon_analysis.product p
join amazon_analysis.order_items o
on p.product_id=o.product_id
group by p.product_category_name
order by total_revenue desc
limit 5;
```

Data Output Messages Notifications

	product_category_name character varying	total_revenue numeric
1	beleza_saude	1257865.34
2	relogios_presentes	1203060.32
3	cama_mesa_banho	1032268.59
4	esporte_lazer	985881.10
5	informatica_acessorios	910605.07

Analysis:

- Lifestyle category and household essentials dominate the total revenue
- Beauty and health winning on revenue= high purchase frequency + decent price point – people reorder these regularly

Recommendations:

- These top 5 categories are priority investment areas for amazon india- deeper inventory, better deals, faster delivery
- Beauty and health are massive and growing market in India (Nykaa's success proves it)
- Prioritize these 5 categories for prime day homepage placement

Analysis-III

1. total sales by seasons –comparing sales between different seasons to spot seasonal trends

```sql
select season,sum(total_sales) as total_sales
from(
select oi.price as total_sales,
       case when extract(month from o.order_purchase_timestamp) in (3,4,5) then 'spring'
            when extract(month from o.order_purchase_timestamp) in (6,7,8) then 'summer'
            when extract(month from o.order_purchase_timestamp) in (9,10,11) then 'autumn'
            else 'winter'
end as season
from amazon_analysis.orders o
join amazon_analysis.order_items oi on o.order_id=oi.order_id
) as sub
group by season
order by total_sales desc;
```

Data Output Messages Notifications

Showing rows: 1 t

	season text	total_sales numeric
1	spring	4216721.54
2	summer	4120359.62
3	winter	2905750.03
4	autumn	2348812.51

Analysis:

- Sales during spring and summer season combined drives over 60% of total revenue
- lowest sales during autumn- no major demand trigger

Recommendations:

- Increase ad campaign & big discounts before spring (educational institutions holidays + summer shopping) season
- autumns weakness = opportunity to create demand through artificial sale events (like amazon great Indian festival fits perfectly in this gap)

2. products above average sales volume – identifying products that have sales volumes above the overall average quantity

```sql
select product_id,count(*) as total_quantity_sold
from amazon_analysis.order_items
group by product_id
having count(*) >
(
select avg(total_quantity)
from
(
select count(*) as total_quantity
from amazon_analysis.order_items
group by product_id
) as product_totals
)
order by total_quantity_sold desc;
```

Data Output Messages Notifications

	product_id character varying	total_quantity_sold bigint
1	aca2eb7d00ea1a7b8ebd4e6831466...	527
2	99a4788cb24856965c36a24e339b6...	488
3	422879e10f46682990de24d770e7f...	484
4	389d119b48cf3043d311335e499d9...	392
5	368c6c730842d78016ad823897a37...	388
6	53759a2ecddad2bb87a079a1f1519f...	373
7	d1c427060a0f73f6b889a5c7c61f2a...	343

Analysis:

- Top 3 products (527, 488, 484) are significantly ahead of the rest
- Most products in 20-100 range are above average and stable, they're worth moving

Recommendations:

- Top 3–5 products need dedicated inventory buffers, priority fulfillment and zero stockout tolerance — losing them hurts badly
- Use this above-average filter as a weekly inventory health check — any previously above-average product dropping below threshold is an early warning signal audit, delist, push sellers to improve

3. monthly revenue trends (2018) – tracking monthly revenue in year 2018 to detect peak and low sales periods

```sql
select round(sum(oi.price),0) as revenue,
       EXTRACT(MONTH FROM o.order_purchase_timestamp) AS month
from amazon_analysis.order_items oi
join amazon_analysis.orders o
on oi.order_id= o.order_id
where o.order_purchase_timestamp between '2018-01-01' and '2018-12-31'
group by month
order by month asc;
```

Data Output Messages Notifications

	revenue numeric	month numeric
1	950030	1
2	844179	2
3	983213	3
4	996648	4
5	996518	5
6	865124	6
7	895507	7

Analysis:

- The revenue is strong and stable across Jan-Aug ranging from 844k-996k
- April and may peak at ~996k the highest months of 2018

Recommendations:

- The APR-MAY is a peak season this is a signal to lock in for a major campaign window for amazon india
- Septembers 145BRL Is a data quality red flag, never use a partial month data in financial reporting

4. customer segmentation by order frequency (CTE) – group customers as occasional, regular, loyal based on their order count

```sql
with customer_segments as (
select o.customer_id,
       count(o.order_id) as total_orders,
       case
       when count( o.order_id) between 1 and 2 then 'occasional'
       when count( o.order_id) between 3 and 5 then 'regular'
       else 'loyal'
     end as customer_type
from amazon_analysis.orders o
group by o.customer_id
)
select customer_type,
       count(*)as count from customer_segments
group by customer_type
order by count desc;
```

Data Output Messages Notifications

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	customer_type text	count bigint
1	occasional	98144
2	regular	106
3	loyal	98

Analysis:

- Amazon brazil is almost one-and-done platform for nearly all users
- Regular + loyal customers combined = 204 out of 98k+ which is about 0.2% retention rate
- brazil is spending heavily on acquisition and getting almost nothing in return

Recommendations:

- Brazils retention failure is biggest lesson.
- Invest in post-purchase experience aggressively — delivery quality, easy returns, personalized follow-ups are what convert Occasional to Regular
- Set clear KPIS 5% of occasional buyers to regular within 12 months

5. Top 20 high value customers – rank customers by their average order value to highlight the most valuable ones

```sql
with cte as(
select
      o.customer_id,round(avg(oi.price),0) as avg_order_value
from amazon_analysis.orders o
join amazon_analysis.order_items oi
on oi.order_id= o.order_id
group by o.customer_id
)
select customer_id,avg_order_value,
       rank() over (order by avg_order_value desc) as customer_rank
from cte
order by customer_rank asc
limit 20;
```

Data Output Messages Notifications

	customer_id character varying	avg_order_value numeric	customer_rank bigint
1	c6e2731c5b391845f6800c97401a4...	6735	1
2	f48d464a0baaea338cb25f816991ab...	6729	2
3	3fd6777bbce08a352fddd04e4a7cc8...	6499	3
4	df55c14d1476a9a3467f131269c24...	4799	4
5	24bbf5fd2f2e1b359ee7de94defc4a15	4690	5
6	3d979689f636322c62418b6346b1c...	4590	6
7	1afc82cd60e303ef09b4ef9837c950...	4400	7

Analysis:

- Top 20 customers have average value 10x than others, these are premium and ultra-premium buyers
- They are high value customers across amazon brazil

Recommendations:

- Build an invite-only tier (like Amazon Prime Platinum) specifically for top avg-order-value customers — exclusive early access, dedicated support, free express delivery
- Use BRL 4,500+ avg order value as the rewards program entry threshold — maps to roughly ₹45,000+ in India, targeting premium electronics and appliance buyers

6. product lifecycle cumulative sales (recursive CTE) – calculate month on month cumulative sales from its first month

```sql
WITH RECURSIVE monthly_data AS (
    SELECT
        oi.product_id,
        DATE_TRUNC('month', o.order_purchase_timestamp) AS sale_month,
        SUM(oi.price) AS monthly_sales
    FROM amazon_analysis.order_items oi
    JOIN amazon_analysis.orders o
    ON oi.order_id = o.order_id
    GROUP BY oi.product_id,
             DATE_TRUNC('month', o.order_purchase_timestamp)
),
first_months AS (
    -- separately find first month per product
    -- no confusion
    SELECT
        product_id,
        MIN(sale_month) AS first_month
    FROM monthly_data
    GROUP BY product_id
),
cumulative AS (
    -- BASE CASE: start from first month only
    SELECT
        md.product_id,
        md.sale_month,
        md.monthly_sales,
        md.monthly_sales AS total_sales
    FROM monthly_data md
    JOIN first_months fm
    ON md.product_id = fm.product_id
    AND md.sale_month = fm.first_month

    UNION ALL
```

```sql
    -- RECURSIVE CASE: keep adding next month
    SELECT
        md.product_id,
        md.sale_month,
        md.monthly_sales,
        ROUND((c.total_sales + md.monthly_sales)::numeric, 2)
    FROM monthly_data md
    JOIN cumulative c
    ON md.product_id = c.product_id
    AND md.sale_month = c.sale_month + INTERVAL '1 month'
)
SELECT
    product_id,
    date(sale_month) as sale_month,
    total_sales
FROM cumulative
ORDER BY product_id, sale_month asc;
```

Data Output Messages Notifications

	product_id character varying	sale_month text	total_sales numeric
1	00066f42aeeb9f3007548bb9d3f33c38	2018-05	101.65
2	00088930e925c41fd95ebfe695fd2655	2017-12	129.90
3	0009406fd7479715e4bef61dd91f2462	2017-12	229.00
4	000b8f95fcb9e0096488278317764d...	2018-08	117.80
5	000d9be29b5207b54e86aa1b1ac54...	2018-04	199.00
6	0011c512eb256aa0dbbb544d8dffcf6e	2017-12	52.00
7	00126f27c813603687e6ce486d909d...	2017-09	498.00
8	001795ec6f1b187d37335e1c470476...	2017-10	38.90
9	001795ec6f1b187d37335e1c470476...	2017-11	116.70

This is **complicated** and does not require recursive CTE as this is a large dataset the recursive data needs to go through 62k+ rows doing cumulative (running total of units sold over a month) sales that consumes a lot of time a heavy load on the database. As an analyst our job is to find the right analysis and we can use the window function approach which is simpler and faster than recursive CTE.

Product life cycle cumulative sales (using window functions)-

```sql
WITH monthly_sales AS (
    SELECT
        oi.product_id,
        DATE_TRUNC('month', o.order_purchase_timestamp) AS sale_month,
        SUM(oi.price) AS monthly_total
    FROM amazon_analysis.order_items oi
    JOIN amazon_analysis.orders o USING (order_id)
    GROUP BY oi.product_id, sale_month
)
SELECT
    product_id,
    TO_CHAR(sale_month, 'YYYY-MM') AS sale_month,
    ROUND(SUM(monthly_total) OVER(
        PARTITION BY product_id
        ORDER BY sale_month
    )::numeric, 2) AS total_sales
FROM monthly_sales
ORDER BY product_id, sale_month asc;
```

Data Output Messages Notifications

Showing rows: 1 to 1

	product_id character varying	sale_month text	total_sales numeric
1	00066f42aeeb9f3007548bb9d3f33c38	2018-05	101.65
2	00088930e925c41fd95ebfe695fd2655	2017-12	129.90
3	0009406fd7479715e4bef61dd91f2462	2017-12	229.00
4	000b8f95fcb9e0096488278317764d...	2018-08	117.80
5	000d9be29b5207b54e86aa1b1ac54...	2018-04	199.00
6	0011c512eb256aa0dbbb544d8dffcf6e	2017-12	52.00
7	00126f27c813603687e6ce486d909d...	2017-09	498.00
8	001795ec6f1b187d37335e1c470476...	2017-10	38.90

Analysis:

- Some products shows consistent growth from 30→350BRL over 8-10months these are our star products -reliable, consistent sellers and other shows exclusive spikes over just 2-4 months these are viral/trending products
- Some products reach cumulative sales of 3000BRL+ in 1-2 months, these huge ticket items needs to be premium placement, Emi options and warranty assurance to keep selling

Recommendations:

- Stock aggressively products showing 3+ consecutive months of growth Bundle slow movers with fast-growing products to boost their sales
- Set up reorder alerts for products showing sudden month-over-month spikes, these go out of stock and lose momentum

7. payment method sales growth (2018) – measure total sales and month-on-month growth rate by payment method.

```sql
WITH monthly_sales AS (
    SELECT
        payment_type,
        DATE_TRUNC('month', order_purchase_timestamp) AS sale_month,
        ROUND(SUM(payment_value)::numeric, 2) AS monthly_total
    FROM amazon_analysis.payments
    JOIN amazon_analysis.orders
    USING (order_id)
    WHERE EXTRACT(YEAR FROM order_purchase_timestamp) = 2018
    GROUP BY payment_type, sale_month
)
SELECT
    payment_type,
    TO_CHAR(sale_month, 'YYYY-MM') AS sale_month,
    monthly_total,
    ROUND((monthly_total - LAG(monthly_total) OVER(
        PARTITION BY payment_type ORDER BY sale_month
    )) * 100.0 / NULLIF(LAG(monthly_total) OVER(
        PARTITION BY payment_type ORDER BY sale_month
    ), 0), 2) AS monthly_change
FROM monthly_sales
ORDER BY payment_type, sale_month;
```

Data Output Messages Notifications

	payment_type character varying	sale_month text	monthly_total numeric	monthly_change numeric
1	boleto	2018-01	204844.66	[null]
2	boleto	2018-02	183112.72	-10.61
3	boleto	2018-03	191538.02	4.60
4	boleto	2018-04	193547.09	1.05
5	boleto	2018-05	195378.93	0.95
6	boleto	2018-06	153350.28	21.51

- **Analysis:**
  the payment type “credit card” tops the monthly growth rate followed by boleto, debit card and vouchers
- Analysis all payment methods growth, steady, declining over the month-on-month and use them to come up with different payment options, marketing ads and campaigns

Recommendations:

- Debit card spike- invest heavily in UPI cashback offers, indias UPI base is 300M+users, potential is enormous
  Boleto/COD- introduce prepaid COD discounts (pay digitally, get 2% extra off)
- Voucher cliff promotions- design multiple vouchers (signup, after first purchase, at 30 days)
- Since the august data is incomplete don't make inventory or budget decisions based on august figures

Thanks & regards
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